Probability

CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course

TODAY

Random Variables

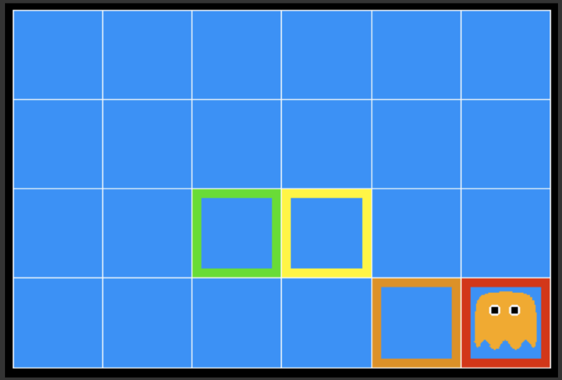
Joint and Marginal Probabilities

Conditional Probabilities

Product Rule, Chain Rule, Bayes Rule

INFERENCE IN GHOSTBUSTERS

A ghost is in the grid somewhere



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Sensor readings tell how close a square is to the ghost

- On the ghost: red
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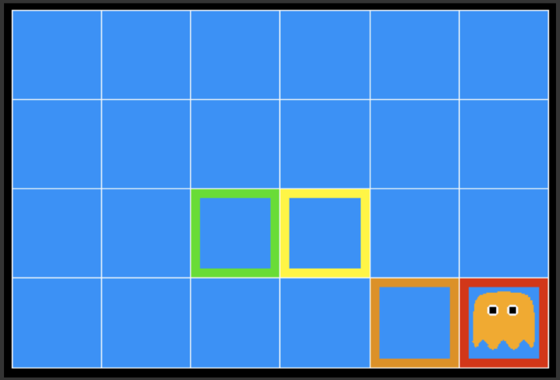

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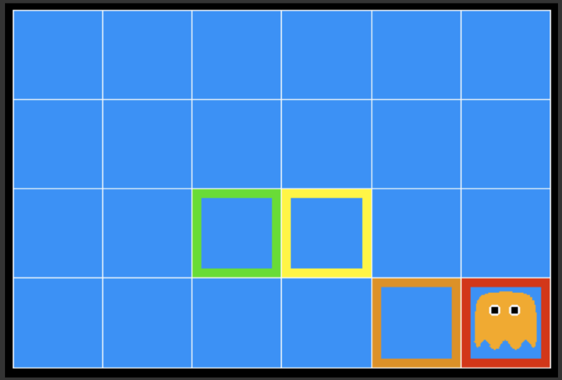

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Sensors are noisy, but we know $P(color|distance)$

$pred(red|3)$	$pred(orange|3)$	$pred(yellow|3)$	$pred(green|3)$
0.05	0.15	0.50	0.30

UNCERTAINTY

General situation:

0.11	0.11	0.11
0.11	0.11	0.11
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0.17	0.10	0.10
0.09	0.17	0.10
0.01	0.09	0.17

0.01	0.01	0.03
0.01	0.05	0.05
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Probabilistic reasoning gives us a framework for managing our beliefs and knowledge

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RANDOM VARIABLES

A random variable is some aspect of the world about which we (may) have uncertainty

- $R =$ Is it raining?
- $T =$ Is it hot or cold?
- $D =$ How long will it take to drive to work?
- $L =$ Where is the ghost?

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Like variables in a CSP, random variables have domains

- $R \in \{false, true\}$ (often written as $\{-r, +r\}$)
- $T \in \{hot, cold\}$
- $D \in [0, \infty)$
- $L \in$ **possible locations**, maybe $\{(0,0), (0,1), \ldots\}$

PROBABILITY DISTRIBUTIONS

Associate a probability with each value

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Temperature $P(T)$

T	P
hot	0.5
cold	0.5

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Temperature $P(T)$

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Weather $P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

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$$\forall x \;\; P(X = x) \geq 0$$

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Shorthand Notation:

$$P(hot) = P(T = hot)$$
$$P(cold) = P(T = cold)$$
$$P(rain) = P(W = rain)$$

OK, *if* all domains entries are unique

JOINT DISTRIBUTIONS

A joint distribution over a set of random variables: $X_1, X_2, \ldots, X_n$

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hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
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$$P(X_1, \ldots, X_n)$$
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Size of distribution if n variables with domain sizes d?

- For all but the smallest distributions, impractical to write out !!

PROBABILISTIC MODELS

A probabilistic model is a joint distribution over a set of random variables

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- (Random) variables with domains
- Assignments are called outcomes
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- Normalized: sum to 1.0
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Constraint satisfaction problems:

- Variables with domains
- Constraints: state whether assignments are possible
- Ideally: only certain variables directly interact

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

T	W	P
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T

EVENTS

An *event* is a set E of outcomes:

$$P(E) = \sum_{(x_1,...,x_n)\in E} P(x_1, \ldots, x_n)$$

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From a joint distribution, we can calculate the probability of any event

- Probability that it is hot AND sunny?
- Probability that it is hot?
- Probability that it is hot OR sunny?

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From a joint distribution, we can calculate the probability of any event

- Probability that it is hot AND sunny?
- Probability that it is hot?
- Probability that it is hot OR sunny?

Typically, the events we care about are partial assignments, like $P(T = hot)$

MARGINAL DISTRIBUTIONS

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cold	rain	0.3

T	P
hot	0.5
cold	0.5

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Marginalization (summing out): Combine collapsed rows by adding

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

T	P
hot	0.5
cold	0.5

W	P
sun	0.6
rain	0.4

CONDITIONAL DISTRIBUTIONS

A simple relation between joint and conditional probabilities

- In fact, this is taken as the definition of a conditional probability

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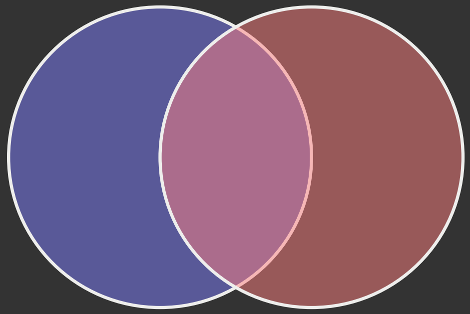

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

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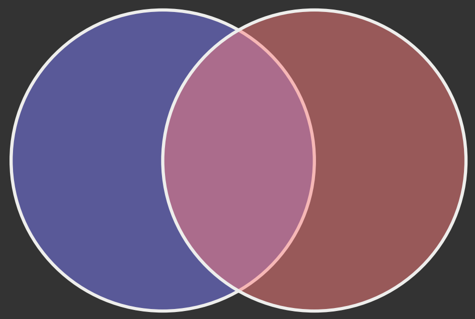

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$$P(W=s|T=c) = \frac{P(W=s,T=c)}{P(T=c)} = \frac{P(W=s,T=c)}{P(W=s,T=c)+P(W=r,T=c)} = \frac{0.2}{0.2+0.3} = 0.4$$

CONDITIONAL DISTRIBUTIONS

Conditional distributions are probability distributions over some variables given fixed values of others

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hot	rain	0.1
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Conditional distributions are probability distributions over some variables given fixed values of others

$$P(W|T = hot)$$

W	P
sun	0.8
rain	0.2

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

CONDITIONAL DISTRIBUTIONS

Conditional distributions are probability distributions over some variables given fixed values of others

$$P(W|T = hot)$$

W	P
sun	0.8
rain	0.2

$$P(W|T = cold)$$

W	P
sun	0.4
rain	0.6

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

NORMALIZATION TRICKS

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W	P
sun	
rain	

NORMALIZATION TRICKS

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$\xrightarrow[\text{matching the evidence}]{\text{SELECT the joint probabilities}}$

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$\xrightarrow[\text{(make it sum to one)}]{\text{NORMALIZE the selection}}$

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NORMALIZATION TRICKS

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SELECT the joint probabilities matching the evidence →

T	W	P
cold	sun	0.2
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NORMALIZE the selection (make it sum to one) →

W	P
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Why does this work?

- Sum of selection is P(evidence)!
- $P(T = c)$ in this case

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- Sum of selection is P(evidence)!
- $P(T=c)$ in this case

In general:

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

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(Dictionary) To bring or restore to a normal condition

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$\xrightarrow[Z = 0.5]{\text{NORMALIZE}}$

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X	Y	P
hot	sun	20
hot	rain	5
cold	sun	10
cold	rain	15

$\xrightarrow[Z = 50]{\text{NORMALIZE}}$

X	Y	P
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- $P(\textbf{on time} | \textbf{no reported accidents}) = 0.90$
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Probabilities change with new evidence:

- $P(\textbf{on time}|\textbf{no accidents, 5 a.m.}) = 0.95$
- $P(\textbf{on time}|\textbf{no accidents, 5 a.m., raining}) = 0.80$
- Observing new evidence causes beliefs to be updated

INFERENCE BY ENUMERATION

Problem Setup:

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- General Case:
  - Evidence variables: $E_1, \ldots, E_k = e_1, \ldots, e_k$
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  - Hidden variables: $H_1, \ldots, H_r$

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Solution:

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Solution:

- Step 1: Select the entries consistent with the evidence
- Step 2: Sum out H to get joint of Query and evidence

$$P(Q, e_1 \dots, e_k) = \sum_{h_1, \dots, h_r} P(Q, h_1, \dots, h_r, e_1, \dots, e$$

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- Step 3: Normalize

$$Z = P(Q, e_1, \ldots, e_r)$$

$$P(Q|e_1 \ldots, e_r) = \frac{1}{Z} P(Q, e_1, \ldots, e_r)$$

INFERENCE BY ENUMERATION

$P(W)$?

$P(W|winter)$?

$P(W|winter, hot)$?

summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

INFERENCE BY ENUMERATION

Obvious problems:

INFERENCE BY ENUMERATION

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- Worst-case time complexity $\mathcal{O}(d^n)$

INFERENCE BY ENUMERATION

Obvious problems:

- Worst-case time complexity $\mathcal{O}(d^n)$
- Space complexity $\mathcal{O}(d^n)$ to store the joint distribution

PRODUCT RULE

Sometimes we have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y)$$
$$\Rightarrow P(x|y) = \frac{P(x, y)}{P(y)}$$

PRODUCT RULE

$$P(y)P(x|y) = P(x, y)$$

Example:

W	P
sun	0.8
rain	0.2

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

CHAIN RULE

More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_2, x_1)$$

$$P(x_1, x_2, \ldots, x_n) = \prod_i P(x_i|x_1, \ldots, x_n)$$

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$$P(x_1, x_2, \ldots, x_n) = \prod_i P(x_i|x_1, \ldots, x_n)$$

This is always true. Why?

BAYES RULE

BAYES RULE

Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

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Dividing, we get:

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- Lets us build one conditional from its reverse
- Often one conditional is tricky but the other one is simple
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- Note: you should still get stiff necks checked out !! Why?

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  - $R =$ reading color measured at $(1, 1)$
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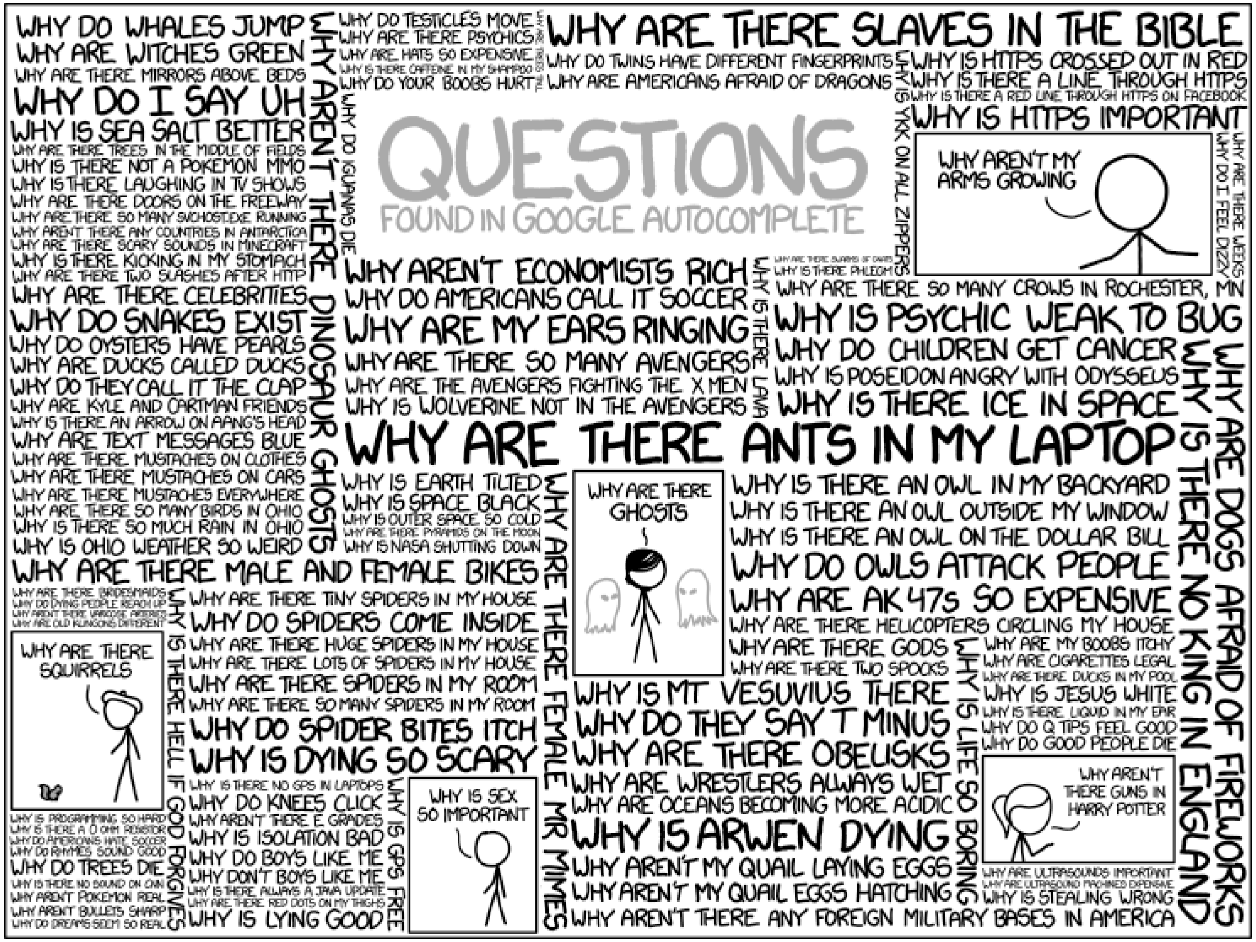
We can calculate the posterior distribution $P(G|r)$ over ghost locations given a reading using Bayes' rule:

$$P(g|r) \propto P(r|g)P(g)$$

Q & A



QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY AREN'T THERE DINOSAUR GHOSTS
WHY DO IGUANAS DIE
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS YKK ON ALL ZIPPERS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS THERE LAVA
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY ARE THERE SQUIRRELS
WHY IS THERE HELL IF GOD FORGIVES
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY IS GPS FREE
WHY IS SEX SO IMPORTANT
WHY DO TREES DIE
WHY ARE THERE FEMALE MR MIMES
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS LIFE SO BORING
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY IS STEALING WRONG
WHY IS THERE NO KING IN ENGLAND
WHY ARE DOGS AFRAID OF FIREWORKS

XKCD